DigiNest

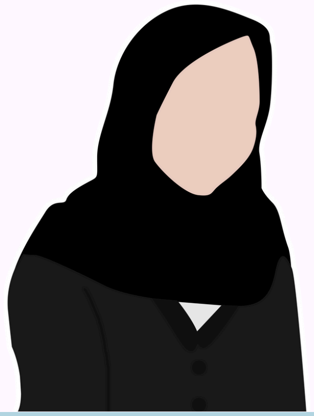
Course Name: System Analysis and Design

Group Name: HallHackers

Making Hall Life Smarter

Date: 19th August 2025

Our Team



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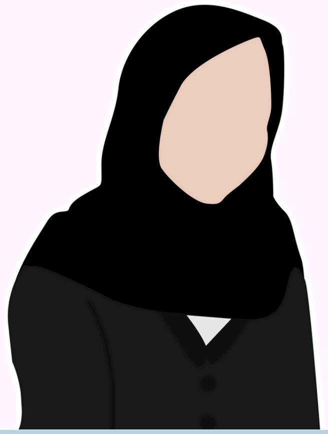

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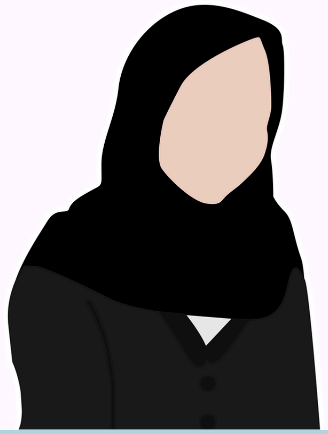

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Project Context

Female Hall Management System

- Female Hall Management System for IUT female students.
- Purpose: Manage student rooms, leave requests, and complaints efficiently.
- Focus: Detailing primitive processes using structured techniques.

Goal: Provide a centralized, automated system for request approvals, complain tracking, and hall facility management.

Selected Primitive Process

- Assign Room to Student(DFD X.X)
- Ensures requests are complete and within rules.
- **Technique:** Structured English (clear sequential steps).
- Approve/Reject Leave Request(DFD X.X)
- Determines complaint validity & escalation rules.
- **Technique:** Decision Table (multiple conditions).
- Route Complaint to Authority(DFD X.X)
- Produces summaries for admins (e.g., complaints/month).
- **Technique:** Decision Tree (branching based on report type).

Process Specification: Assign Room

Process ID:	1.1
Name:	Assign Room to Student
Inputs:	Inputs: StudentID, Requested Room Type, Available Rooms
Outputs:	Assigned Room Number, Room Assignment Confirmation, Waiting List Update, Notification.
Description:	Assigns an available room to a student or adds them to the waiting list if unavailable.
Technique:	Structured English

Room Assignment Logic: (Stuctured English)

```
IF AvailableRooms for Requested Room Type is empty THEN
   REJECT "Requested room type not available"
   ADD StudentID to waiting list
   NOTIFY student about unavailability
ELSE
   ASSIGN first available room to StudentID
   UPDATE RoomStatus to "Occupied"
   GENERATE RoomAssignmentConfirmation for StudentID
END
```

Room Assignment Logic: (Stuctured English)

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```

Process Specification: Approve/Reject Leave Request

Specification Form	
Process ID:	1.2
Name:	Fee Payment
Inputs:	StudentID, Amount Paid, Payment Date, Payment Mode, Scholarship Status, Hall Eligibility List
Outputs:	Payment Status (Accepted / Rejected / Not Required / Late Fee Applied), Notification to Student
Description:	checks whether the student is eligible for the hall, verifies scholarship status, validates payment amount and timeliness, and updates the system.
Technique:	Decision Table

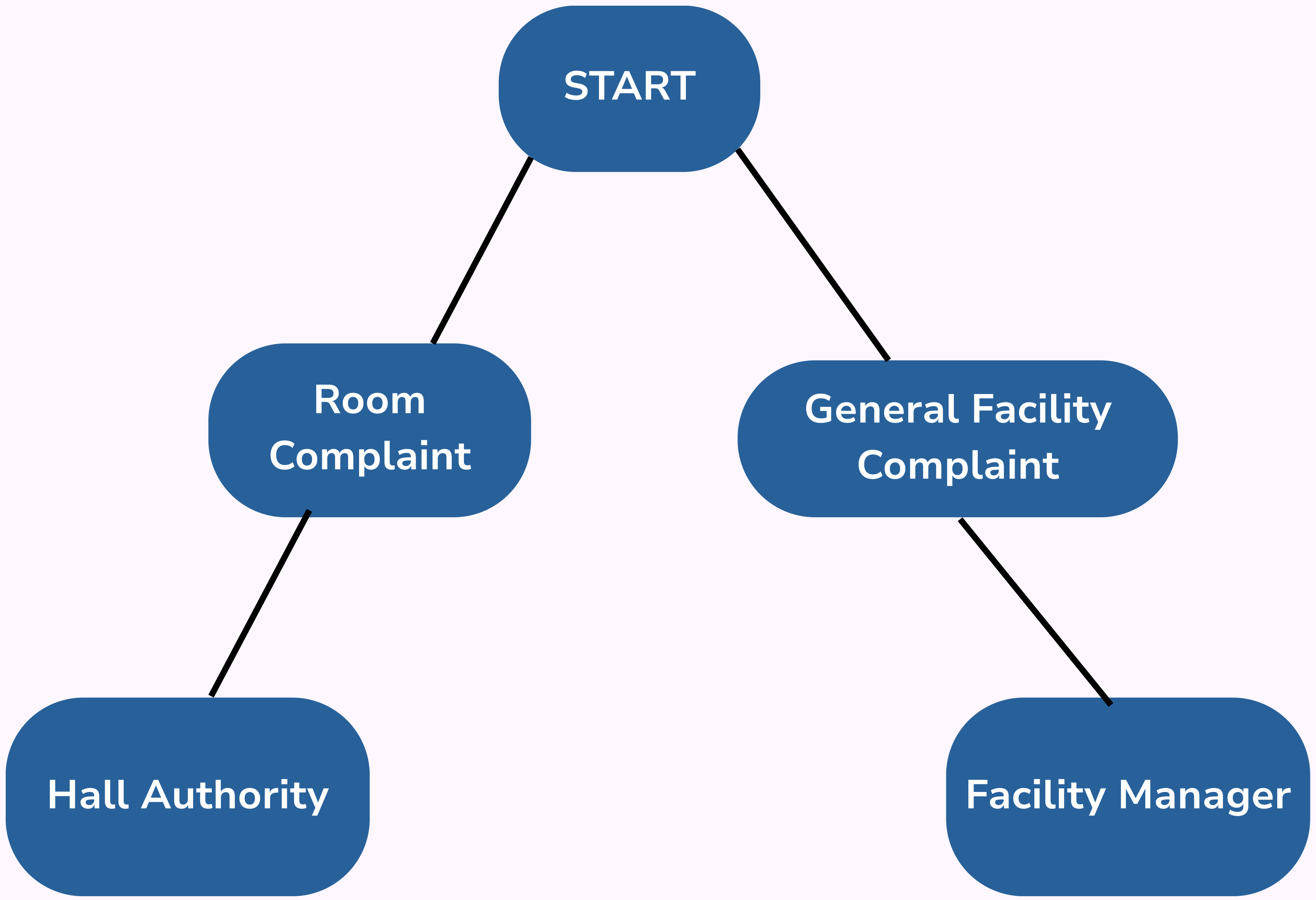
Decision Table

Rule	Amount Paid Correct?	On Time?	Scholarship?	On Hall List?	Actions
R1	-	-	Yes	Yes	Payment Not Required
R2	Full	Yes	No	Yes	Accept Payment
R3	Full	Late	No	Yes	Reject Payment + Notify Student
R4	Incorrect	-	No	Yes	Reject Payment + Notify Student
R5	Paid (shouldn't)	-	Yes	No	Refund Payment + Notify Student

Process Specification:
Route Complaint to Appropriate Authority

Specification Form	
Process ID:	1.3
Name:	Route Complaint to Appropriate Authority
Inputs:	StudentID, Complaint Type (Personal Room / General Facility).
Outputs:	Assigned Authority (Maintenance Staff, Cleaning Staff, Facility Manager), Notification to Authority
Description:	Determines the correct authority based on the nature of the complaint and routes it accordingly for efficient resolution.
Technique:	Decision Tree

Complaint Handling Decision Tree



Process Specification

Process: Assign Room to Student

- **Inputs:** StudentID, RoomPreference, Availability
- **Outputs:** Assigned Room Number / Waiting List Entry.
- **Why Primitive?** Only assignment logic, no further breakdown.
- **Technique:** Structured English.

Thank You

We are now ready to address any questions or clarifications